

ENGE707 – Data Engineering and AI

Contribution Statement for Phase 2

Group (iRobot)

Collaboration

All team members contributed to planning Phase 2 and agreeing the shared experimental protocol, including the fixed train-test split, the evaluation metrics, and the approach to class imbalance. Work was then divided by task, with weekly meetings throughout the phase so members could report progress, raise blockers, and review each other's work. All code was developed in a shared GitHub repository, with individual branches merged into the main branch, so that contributions are visible in the commit history.

Task Allocation:

Task	Contributor(s)
Task 1: Project Phase I Summary	Aman Mohammed
Task 2: Original LCS System on Raw Dataset	Simon Aung
Task 3: Data Preprocessing and Feature Engineering	Aman Mohammed
Task 4: Improved LCS-Based System	Simon Aung
Task 5: Experiments	Khlaif Alshammari
Task 6: Model Comparison	Khlaif Alshammari
Task 7: Interpretation of Results	Khlaif Alshammari
Task 8: Discussion, Limitations, and Future Work	Aman Mohammed
Final report compilation and review	Shared between all group member (each group member worked on the task allocated to them)

Individual Contribution

Aman Mohammed:

My main contributions were Tasks 1, 3, and 8. For Task 1, I summarised the Phase I project, including the dataset, prediction objective, data quality findings, and suitability of an LCS based approach. For Task 3, I completed the data preprocessing and feature-engineering work, including data validation, the reproducible train-test split, class balancing of the training data, and preparation of the datasets required for modelling.

I also completed Task 8 by discussing the reasons behind the model's performance, the practical meaning of the results, the project's limitations, and suitable directions for future work. I reviewed my sections against the notebook results and contributed to compiling the final report.

Khlaif Alshammari:

My contribution to the project was completing Tasks 5, 6 and 7. For Task 5, I designed the experimental setup, including the stratified train-test split, the ten-fold cross-validation used within the training data, and the selection of evaluation metrics suitable for the imbalanced target. I built a shared evaluation module that fixed the split, the folds and the metric definitions, so that every model in the project was scored by the same code on the same records. I applied the Friedman test, Wilcoxon signed-rank tests and McNemar's test to establish which differences between models were statistically significant and interpreted what those results meant in practice.

For Task 6, I implemented the three conventional machine learning models under three class-imbalance strategies and assembled the full comparison of twelve models, including the original and improved eLCS systems. This showed which imbalance strategy performed best and how the LCS compared with the conventional models. For Task 7, I analysed the rule population produced by the improved eLCS, translated individual classifiers into readable rules, examined which features the system relied on, and assessed whether the explanations agreed with established diabetes risk factors. I also produced the related tables and figures and reviewed my sections against the notebook results.

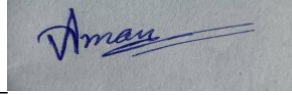
Simon Aung:

My contribution to the project was completing Task 2 and Task 4. For Task 2, I implemented the original eLCS system on the minimally processed diabetes dataset, recorded its parameters and evaluated its baseline performance using accuracy, balanced accuracy, precision, recall, F1-score and a confusion matrix. For Task 4, I developed the improved eLCS system by using balanced training data, defining the feature types and testing different population sizes and `p_spec` values. I selected the best configuration using validation balanced accuracy and evaluated the final model on the unchanged test set. I also documented the method and results, created the relevant graphs and tables, and uploaded my notebook contributions to the team's GitHub repository.

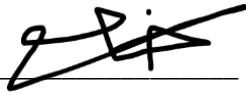
Declaration

We confirm that the above accurately reflects each member's contribution to this phase of the project, and that all members were actively involved in regular team meetings.

Aman Mohammed



Khlaif Alshammari



Simon Aung



Date: 03/10/2026